MUHAMMED ZAKI C T

MERN Stack Developer | Full Stack Software Engineer

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Professional Summary

Full Stack Developer with hands-on experience building responsive web applications using the MERN stack (MongoDB, Express.js, React.js, and Node.js), Next.js, Angular, and modern JavaScript/TypeScript. Skilled in developing REST APIs, reusable UI components, authentication and role-based access control, database modeling, and state management with Redux Toolkit. Also experienced in computer vision workflows using Python, OpenCV, and YOLOv8 for vehicle detection, tracking, and speed estimation. Familiar with Git-based collaboration, Agile workflows, and deploying web applications.

Professional Experience

MERN Full Stack Development Trainee, *Luminar Technolab* 05/2026 – Present | Kozhikode, Kerala

- Develop full-stack web applications using MongoDB, Express.js, React.js, and Node.js, with additional work in Angular and Next.js.
- Build and document REST APIs with Express.js and Node.js, implementing JWT authentication, bcrypt password hashing, and role-based access control.
- Create modular, reusable, accessible, and responsive UI components using React.js, Angular, Tailwind CSS, and modern CSS.
- Design MongoDB schemas and use Mongoose for validation, CRUD operations, and data querying.
- Manage client-side state and asynchronous API calls with Redux Toolkit and React Context API.
- Use Git and GitHub for version control and collaborate within development workflows.

AI and Computer Vision Research Intern 06/2025 – 12/2025 | Kozhikode, Kerala

NIT Calicut – Department of Civil Engineering

- Contributed to an AI-powered intelligent traffic management system using aerial drone footage for vehicle detection, tracking, and speed estimation.
- Processed video streams with Python and OpenCV and prepared custom vehicle datasets using Roboflow.
- Trained and evaluated YOLOv8 object-detection models using PyTorch and Ultralytics for multi-class vehicle detection.
- Developed speed-estimation and traffic-analysis workflows to support transportation monitoring and road-safety analysis.

Technical Skills

Programming Languages
JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, SQL

Backend
Node.js, Express.js, REST APIs, Middleware, JWT Authentication, Password Hashing, Role-Based Access Control (RBAC), MVC

Tools and Platforms
Git, GitHub, Postman, Linux/Bash, npm, yarn, Google Cloud Platform (GCP), Vercel, Roboflow, VS Code

Core Competencies
API Integration, Component-Based UI, State Management, SDLC, Agile/Scrum, Problem Solving

Frontend
React.js, Next.js, Angular, Redux Toolkit, React Router, Tailwind CSS, Bootstrap, Material UI, Vite, Responsive Web Design

Databases
MongoDB, Mongoose ODM, NoSQL Data Modeling, Schema Design, Aggregation Framework, Query Optimization

AI and Computer Vision
OpenCV, YOLOv8, PyTorch, Real-Time Video Analytics, Object Detection and Tracking

Projects

Book Store – Full Stack MERN E-Commerce Application Present

Tech Stack: MongoDB, Express.js, React.js, Node.js, Tailwind CSS, JWT, Google Cloud Platform

- Built a full-stack bookstore application with book catalog browsing, book details, user profile management, and checkout-related workflows.
- Implemented JWT-based authentication and protected routes on the client and server.
- Developed REST API endpoints and MongoDB schemas with Mongoose to support CRUD operations.

Redux Kart – E-Commerce Web Application 2026

Tech Stack: React.js, Redux Toolkit, JavaScript, Tailwind CSS, DummyJSON REST API, Axios

- Built a responsive e-commerce application using Redux Toolkit for centralized state management.
- Implemented product search, category filtering, add-to-cart, quantity updates, subtotal calculation, and item removal.
- Integrated REST APIs with Axios to retrieve and display product data.

Product Management System – Inventory Web Application 2026

Tech Stack: React.js, JavaScript, JSON Server, Axios, React Hooks, Tailwind CSS

- Developed an inventory management application supporting CRUD operations for product name, category, price, stock quantity, description, and image.
- Used React Hooks for state updates, rendering, and REST API communication.

Interactive Resume Builder 2026

Tech Stack: React.js, Vite, Material UI, Axios, JSON Server, Client-Side PDF Generation

- Created a single-page application for creating, editing, customizing, and previewing resumes with live updates.
- Implemented CRUD operations using a JSON Server backend and enabled client-side PDF export.

Spider-Man – Cinematic Interactive Web Application 2026

Tech Stack: React.js, Vite, Tailwind CSS, JavaScript, Scroll-Driven Animation, Vercel

- Created an interactive web experience with scroll-driven animation across approximately 300 visual frames.
- Added interactive carousels, custom cursor behavior, and micro-interactions; deployed on Vercel.

ML-Based Smart Traffic Management System 06/2025 – 12/2025

Tech Stack: Python, YOLOv8, OpenCV, PyTorch, Roboflow, Computer Vision

- Built a computer vision pipeline for vehicle detection, tracking, and speed estimation from aerial drone video.
- Trained custom YOLOv8 model weights for vehicle classification and traffic-flow analysis.

Education

Bachelor of Science (B.Sc.) in Physics, *MAMO College*
06/2023 – 03/2026 | Kozhikode, Kerala

Higher Secondary Education (HSE) – Computer Science,
Rahmaniya HSS
06/2021 – 03/2023 | Kozhikode, Kerala

Languages

English
Professional Working Proficiency

Malayalam
Native Proficiency